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SSA Sharing & Orbital Data Requests (ODR)

[SSA Sharing & Orbital Data Requests](#)[Data Examples & Forms](#)

As the United States government agency responsible for Space Situational Awareness (SSA) information, United States Space Command (USSPACECOM), is committed to promoting a safe, stable, sustainable, and secure space environment through SSA information sharing. As more countries, companies, and non-governmental organizations field space capabilities and benefit from the use of space systems, it is in our collective interest to act responsibly and to enhance overall spaceflight safety. To achieve effective SSA, USSPACECOM seeks to increase cooperation and collaboration with partners and space-faring entities through the exchange of SSA data and provision of SSA services.



The Space Forces - Space (S4S) - Combined Joint Force Space Component Commander (CJFSCC) provides SSA services through this website and the 18th Space Defense Squadron (18 SDS) (<https://www.peterson.af.mil/About/Fact-Sheets/Display/Article/2356622/18th-space-control-squadron/>), located at Vandenberg Space Force Base, California. There are three levels of SSA services: basic, emergency, and advanced.

▼ Basic Services

▼ Basic SSA Information

USSPACECOM provides space surveillance data to registered users through this public website, www.space-track.org. 18 SDS routinely updates the website with positional data on more than 16,000 satellites in orbit around the Earth. Users can build customized API queries (<https://www.space-track.org/#/queryBuilder>) to pull specific data from historical records, and automate the retrieval of new data.

Available products:

- Satellite Catalog (SATCAT) (<https://www.space-track.org/#/catalog>) information
- General Perturbation (GP) positional data for the unclassified space catalog (two line element sets) (<https://www.space-track.org/#/tle>)
- Satellite decay and reentry (<https://www.space-track.org/#/decay>) predictions

Required from customer:

- Valid Space-Track.org user account (<https://www.space-track.org/auth/createAccount>)

▼ Redistribution of Basic SSA Information

USSPACECOM has provided express blanket approval for transfer/redistribution of basic SSA data and services accessed via www.Space-Track.org conditioned on appropriate citation. Publications of analysis based on USSPACECOM data also require appropriate citations.

Basic SSA data are Two-Line Elements (TLEs) and Orbital Mean-element Messages (OMMs); SATCAT; and Satellite Decay and Reentry Data.

Basic SSA data can be found on the Home panel (<https://www.space-track.org/>):

- ELSET Search (<https://www.space-track.org/#/gp>)
- SATCAT (<https://www.space-track.org/#catalog>)
- Decay/Reentry (<https://www.space-track.org/#decay>)

▼ Emergency Services

18 SDS provides emergency services to customers with specific needs, such those who operate satellites.

▼ Anomaly Resolution

Anomaly Resolution consists of actions taken by an owner/operator in response to a launch or on-orbit anomaly or non-nominal condition affecting a spacecraft bus, telemetry subsystem or communications payload, onboard sensor(s), or any other operating payload or subsystem.

Available products:

- Special Perturbation (SP) state vectors (<https://www.space-track.org/documents/State%20Vector%20Example.docx>)



- Unclassified imagery analysis, if available and approved for release

Required from customer:

- Immediately contact 19SDS.orbital.safety@spaceforce.mil (mailto:19SDS.orbital.safety@spaceforce.mil) with the following:
 - Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
 - Details of spacecraft status (e.g., attitude determination, spacecraft configuration, communication status)

▼ Basic Emergency Conjunction Assessment (On-Orbit)

On-Orbit Conjunction Assessment (CA) is the process for determining the point and time of closest approach of two tracked orbiting objects. 18 SDS screens all active satellites against the satellite catalog several times per day to identify close approaches. If a close approach meets emergency reportable criteria (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf), 18 SDS will notify the satellite's owner/operator by email and through Space-Track.org's Operator Panel. This service is provided at no cost to promote spaceflight safety, and does not require an SSA Sharing Agreement or an ODR.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf)
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) on Space-Track.org

Required from customer:

- Contact SPOC.SPACE.DataSharing@spaceforce.mil (mailto:SPOC.SPACE.DataSharing@spaceforce.mil) and provide:
 - Contact information: email addresses and phone number(s)
 - Space-Track.org usernames
 - The catalog number(s) and name(s) of the satellites you own or operate

▼ Basic Emergency Collision Avoidance (On-Orbit)

If a satellite owner/operator receives a close approach notification email from 18 SDS, the owner/operator may submit their proposed maneuver plan and ephemeris for screening against the space catalog. Results that meet emergency reportable criteria (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) will be provided. Collision avoidance is the responsibility of the satellite owner/operator, but 18 SDS will support their collision avoidance efforts by screening their predictive ephemeris and providing results in the Conjunction Data Message (CDM) format.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf)
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) on Space-Track.org

Required from customer:

- Predictive ephemeris formatted and named in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)



- Notification of any planned, confirmed, or cancelled maneuvers, in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)

▼ Advanced Services

Advanced services are available to all entities who sign an SSA Sharing Agreement with USSPACECOM.

▼ Launch Conjunction Assessment

Launch Conjunction Assessment (CA) identifies potential conjunctions that may result in a collision between launching objects and on-orbit objects. Using information provided by the customer prior to launch, via the Form 22 (https://www.space-track.org/documents/LCA_Form_22.docx), 18 SDS screens the launch vehicle against the space catalog and provides conjunction information for each associated launch time. Screening begins at an altitude of 150km or greater and continues through the duration of the launch trajectory (https://www.space-track.org/documents/LCA_Trajectory_Formats.docx) provided by customer. The Launch CA screening results identify periods during the launch window which may put the rocket and payload at increased risk for collision.

Available products:

- Launch CA screening results: summary of predicted conjunction time windows, including:
 - Time of launch
 - Catalog number and common name of the conjuncting object
 - Mission Elapsed Time (MET) after liftoff (in seconds) of the predicted close approach
 - Predicted overall miss distance (in km) for the close approach at the MET

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Form 22 (https://www.space-track.org/documents/LCA_Form_22.docx)
- Properly formatted launch trajectory data in EFG coordinate system (https://www.space-track.org/documents/LCA_Trajectory_Formats.docx)

▼ Launch Early Orbit Determination

Early Orbit Determination (EODET) is the process by which the Space Surveillance Network (SSN) tracks a launch and/or separation of an object and sends observation data to 18 SDS for the calculation of initial orbit(s), entry into the space catalog, identification and naming of the object(s), and continual generation of element sets. 18 SDS uses information provided by the launch entity or satellite owner/operator to inform the tracking, cataloging, and identification process.

Available products/services:

- Two-Line Element Sets (<https://www.space-track.org/#/recent>)
- Satellite naming (for satellite owners/operators)

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Launch sequence and orbital parameters



- R-15 form (<https://www.space-track.org/documents/R-15%20Form.docx>) (preferably 30 days prior to launch)
- Satellite registration by satellite owners/operators

▼ Early Orbit Conjunction Assessment

Early Orbit Conjunction Assessment (CA) includes the screening of owner/operator-provided ephemeris against the satellite catalog to facilitate the safe maneuvering of a newly launched object into its final orbit. Expanded screening volumes and propagation (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) are used to identify possible close approaches. Operators may provide more than one ephemeris file for each maneuver to allow analysis of multiple scenarios.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf) for conjunctions that meet emergency reportable criteria
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) for all results within screening volumes

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Predictive ephemeris formatted and named in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Space-Track.org usernames
- R-15 form (<https://www.space-track.org/documents/R-15%20Form.docx>) (optional, but highly encouraged)

▼ Advanced Conjunction Assessment (On-Orbit)

Advanced On-Orbit CA uses expanded screening volumes and propagation (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) to provide more results than Basic Emergency CA.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf) for conjunctions that meet emergency reportable criteria
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) on Space-Track.org for all results within screening volumes

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Predictive ephemeris formatted and named in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) (optional, but highly encouraged)
- Space-Track.org usernames



▼ **Advanced Collision Avoidance (On-Orbit)**

Advanced Collision Avoidance may be provided to assist owners/operators in relocating a satellite, or to avoid self-identified close approaches. Both situations allow an owner/operator to submit maneuver and non-maneuver ephemeris for expanded screening (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) against the satellite catalog.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf) for conjunctions that meet emergency reportable criteria
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) for all results within screening volumes

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Maneuver plan formatted in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Predictive ephemeris formatted and named in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Space-Track.org usernames

➤ **Disposal/End-of-Life Support**

▼ **Deorbit and Reentry Support**

Deorbit and Reentry Support includes reentry assessment, reentry confirmation, and assistance in deorbit operations. Reentry assessments are predictions of the time and location where an uncontrolled object will reenter the atmosphere (not where the object will impact the ground). If a satellite owner/operator decides to perform the controlled deorbit of a satellite or rocket stage and provides relevant on-orbit information, 18 SDS will provide related conjunction assessment and collision avoidance support using expanded screening volumes (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf) and propagation., and confirmation of reentry after the owner/operator's deorbit procedures are performed.

Available products:

- Close Approach Notification email (https://www.space-track.org/documents/CAN_Example.pdf) for conjunctions that meet emergency reportable criteria
- Conjunction Data Messages (CDMs) (https://www.space-track.org/documentation#odr-examples_forms) for all results within screening volumes
- State vectors (<https://www.space-track.org/documents/State%20Vector%20Example.docx>)
 - 1 per day for controlled deorbit/reentry
 - 1 per 8 hours for uncontrolled reentry



- Tracking and Impact Prediction (<https://www.space-track.org/documents/TIP%20message%20example.pdf>) (TIP) messages via Space-Track.org
- Reentry confirmation via Space-Track.org, or by email/phone

Required from customer:

- Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx)
- Maneuver plan formatted in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Predictive ephemeris formatted and named in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Space-Track.org usernames

▼ SSA Sharing Agreement

USSPACECOM/J5 Strategy, Plans, and Policy negotiates SSA Sharing Agreements. The SSA Sharing Agreement establishes the parameters within which data will be exchanged by both signing parties to facilitate ongoing cooperation and advance spaceflight safety. All members of the space community, including satellite operators, launching agencies, commercial service providers, and research/academic institutions, are welcome to contact SPOC.SPACE.CustomerService@spaceforce.mil (mailto:SPOC.SPACE.CustomerService@spaceforce.mil) for more information on the agreement process. U.S. government (USG) organizations and their contractors have implied agreements and do not need to pursue additional documentation with USSPACECOM.

▼ Orbital Data Request

All customers (non-U.S. government, commercial, and USG) should request advanced services using the Orbital Data Request (https://www.space-track.org/documents/USSPACECOM_ODR.docx) (ODR) form. To submit an ODR, follow the directions on page 1 of the form and send it to SPOC.SPACE.DataSharing@spaceforce.mil (mailto:SPOC.SPACE.DataSharing@spaceforce.mil).

18 SDS's SSA Sharing Team processes all requests, which undergo security, legal, and resource assessments before the final 18 SDS authority approves or disapproves the request. In the event of disapproval, the SSA Sharing Team will work with the customer to revise their request for reconsideration. If a request is approved by 18 SDS, but the customer is non-USG and does not have an SSA Sharing Agreement, the request will be forwarded to USSPACECOM for further review.

After approval, the SSA Sharing Team will provide an estimate of when the data will be delivered, or when the ongoing service will be initiated. In some situations this will take more time than others, but the customer will be notified accordingly. ODRs are valid for one year from the date of approval.

Customers may request information not included in standard services, but this will lengthen the review process and the time required to deliver the data or service, if the request is approved.

▼ Register Your Satellite/Payload with Delta 2 SPACE Office



Whether you are putting your first satellite on orbit, or adding to an existing constellation, we encourage you to register (https://www.space-track.org/documents/New_Satellite_Registration_Form.docx) your asset with 18 SDS so that we can plan for optimal tracking and identification, and provide you with conjunction assessment services as soon as possible. As soon as you register, a member of 18 SDS will contact you to discuss the details of your mission and coordinate conjunction assessment and other required support.

▼ Communicating and Coordinating with 18 SDS

Due to the dynamic nature of space operations, 18 SDS relies on information from operators, launching agencies, and the overall space community to maintain space situational awareness. Even if you do not require direct support from 18 SDS, we welcome the following information:

- Launch information in the R-15 form (<https://www.space-track.org/documents/R-15%20Form.docx>)
 - Complete to the fullest extent possible and email to 18SPCS.doo.launch@us.af.mil (<mailto:18SPCS.doo.launch@us.af.mil>)
- Updates on satellite health and status
 - Satellite breakups, malfunctions, loss of communications, shedding, change of operational status, decommissioning, etc. Email updates to SPOC.SPACE.DataSharing@spaceforce.mil (<mailto:SPOC.SPACE.DataSharing@spaceforce.mil>)
 - If there is an emergency situation, requiring anomaly support, please email 19SDS.orbital.safety@spaceforce.mil (<mailto:19SDS.orbital.safety@spaceforce.mil>)
- Satellite identification
 - Send identifications and supporting information to SPOC.SPACE.DataSharing@spaceforce.mil (<mailto:SPOC.SPACE.DataSharing@spaceforce.mil>)
- Notification of any planned, confirmed, or cancelled maneuvers, in accordance with guidance in the Spaceflight Safety Handbook for Operators (https://www.space-track.org/documents/SFS_Handbook_For_Operators_V1.7.pdf)
- Positional corrections for satellites as element sets or ephemeris
 - Send TLEs and/or ephemeris to SPOC.SPACE.DataSharing@spaceforce.mil (<mailto:SPOC.SPACE.DataSharing@spaceforce.mil>)
- Catalog corrections
 - Send the correction and supporting information to SPOC.SPACE.DataSharing@spaceforce.mil (<mailto:SPOC.SPACE.DataSharing@spaceforce.mil>)
 - 18 SDS will investigate suggested corrections, but cannot guarantee that changes will be made

Maintaining spaceflight safety and space situational awareness is a cooperative activity and responsibility. The information you provide advances these goals and ensures equal access to space for all global users. If you have questions about USSPACECOM services or suggestions on how we can improve any aspect of our SSA Sharing process, we encourage you to contact the 18 SDS SSA Sharing Team at SPOC.SPACE.DataSharing@spaceforce.mil (<mailto:SPOC.SPACE.DataSharing@spaceforce.mil>).

▼ U.S. Government Entities and Contractors



USG entities and their contractors may be eligible to receive additional data and services not listed above depending on their mission requirements and security clearance. For more information, please email SPOC.SPACE.OrbitalDataRequest@spaceforce.mil (mailto:SPOC.SPACE.OrbitalDataRequest@spaceforce.mil).

▼ **Frequently Asked Questions**

- What do you do with the information provided in the R-15 form?
- Can I submit the R-15 if I don't have all of the information?
- If I don't have an SSA Sharing Agreement can I submit an ODR?
- How long does it take an ODR to be approved?
- I'm a satellite operator who receives CDMs. Do I need to submit an ODR to share these with other organizations?
- My organization just launched a satellite, but it hasn't been identified in the catalog.

What can I do?

▼ **Key Contacts**

Delta 2 SSA Sharing Team

The SSA Sharing Team will respond to your inquiry, or put you in contact with the appropriate 18 SDS personnel.

Email: SPOC.SPACE.DataSharing@spaceforce.mil
(mailto:SPOC.SPACE.DataSharing@spaceforce.mil)

Phone: +1-805-606-2675 or DSN 276-2675

19 SDS Orbital Safety Team

Please contact 19 SDS Orbital Safety Team for routine conjunction assessment support. They are manned 24 hours a day, 7 days a week.

Email: 19SDS.orbital.safety@spaceforce.mil (mailto:19SDS.orbital.safety@spaceforce.mil)

Phone: +1-540-284-3999 or DSN 284-3999

